

D.3 Motion in electromagnetic fields

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **24 marks**

$e = 1.6 \times 10^{-19} \text{ C}$; $m_e = 9.11 \times 10^{-31} \text{ kg}$. Show your working.

Question 1 — Aurora australis

(9 marks)

High above the South Island, an electron of the solar wind moves at $2.0 \times 10^7 \text{ m s}^{-1}$ perpendicular to the Earth's magnetic field of strength $5.0 \times 10^{-5} \text{ T}$.

- (a) State the expression for the magnetic force on a moving charge, and the geometrical condition for the force to be maximum. [2]
- (b) Calculate the force on the electron. [2]
- (c) Explain why this force changes the electron's *direction* but never its *speed*. [3]
- (d) State the shape of the electron's path while the field is uniform and the velocity stays perpendicular to it. [2]

Question 2 — The particle merry-go-round

(7 marks)

For a charged particle moving perpendicular to a uniform magnetic field, the magnetic force provides the centripetal force.

- (a) Show that the radius of the path you described in Question 1(d) is $r = \frac{mv}{qB}$. [3]
- (b) Calculate the radius for the electron of Question 1. [2]
- (c) A proton ($m_p \approx 1836 m_e$) enters the same field at the same speed. State how its radius compares, and why the aurora's structure cares about particle mass. [2]

Question 3 — The velocity bouncer

(6 marks)

A velocity selector admits charged particles into a region where an electric field $E = 3.0 \times 10^4 \text{ V m}^{-1}$ and a magnetic field $B = 1.5 \times 10^{-2} \text{ T}$ are crossed (perpendicular to each other and to the beam), arranged so their forces on a moving charge oppose.

- (a) Show that the speed that passes through undeflected is $v = E/B$, and calculate it. [3]
- (b) State what happens to particles moving faster than this speed, and why. [2]
- (c) Explain why the selected speed does not depend on the sign of the particle's charge. [1]

Question 4 — Riding the field line

(2 marks)

Another electron finds itself moving exactly *parallel* to the magnetic field.

- (a) State the magnetic force on it, and hence describe its motion. [2]

Answers

Question 1 — Aurora australis

- (a) $F = qvB \sin \theta$, where θ is the angle between the velocity and the field. The force is maximum when the velocity is *perpendicular* to the field, $\theta = 90^\circ$.
- (b) $F = qvB = (1.6 \times 10^{-19})(2.0 \times 10^7)(5.0 \times 10^{-5}) = 1.6 \times 10^{-16} \text{ N}$.
- (c) The magnetic force is always perpendicular to the velocity, so it does no work on the electron: $W = Fs \cos 90^\circ = 0$. No work means no change in kinetic energy, so the *speed* is constant. What the force does change is the *direction* of the velocity, and since the force stays perpendicular to a velocity of unchanging magnitude, it acts purely as a centripetal force.
- (d) A circle: uniform circular motion at constant speed.

Question 2 — The particle merry-go-round

- (a) The magnetic force supplies the centripetal force:

$$qvB = \frac{mv^2}{r} \Rightarrow qB = \frac{mv}{r} \Rightarrow r = \frac{mv}{qB}$$

(b) $r = \frac{(9.11 \times 10^{-31})(2.0 \times 10^7)}{(1.6 \times 10^{-19})(5.0 \times 10^{-5})} = 2.3 \text{ m}$.

- (c) Since $r \propto m$ at fixed v , q and B , the proton's radius is about 1836 times larger, roughly 4 km rather than a couple of metres. Mass therefore sorts the incoming particles into vastly different circles, which is why the aurora's fine structure depends on which particles arrive.

Question 3 — The velocity bouncer

- (a) Undelected means the electric and magnetic forces balance: $qE = qvB$, so $v = \frac{E}{B} = \frac{3.0 \times 10^4}{1.5 \times 10^{-2}} = 2.0 \times 10^6 \text{ m s}^{-1}$.
- (b) For faster particles the magnetic force qvB grows with v while the electric force qE does not, so the magnetic force wins and the net force deflects them off the axis. They strike the walls and are filtered out.
- (c) The charge q appears in *both* forces and cancels in $qE = qvB$. Reversing the sign of the charge reverses both forces together, so they still balance, and at the same speed.

Question 4 — Riding the field line

- (a) With the velocity parallel to the field, $\theta = 0$ and $F = qvB \sin 0 = 0$: there is no magnetic force at all. The electron therefore continues in a straight line at constant velocity. (A helical path arises only when the velocity has a component perpendicular to the field as well.)